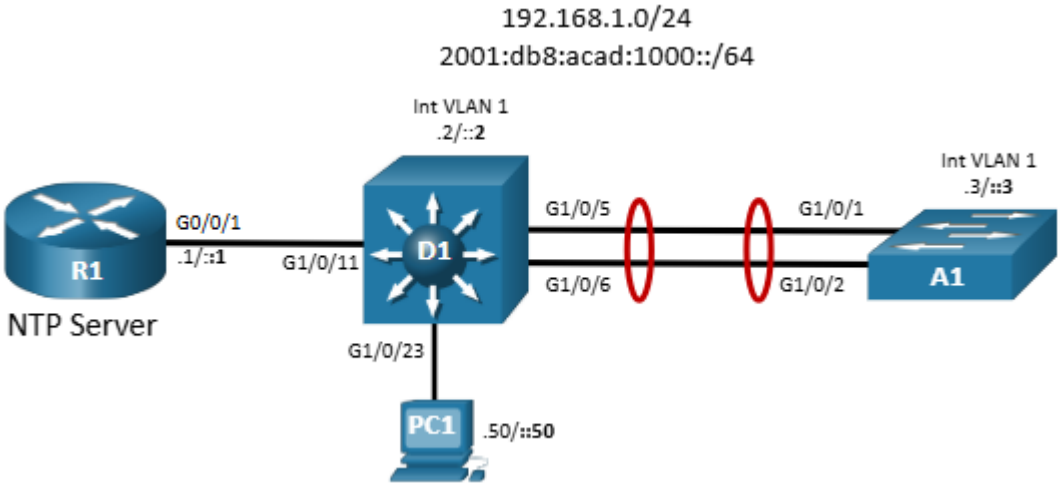
Lab - Implement SNMP and Syslog

Topology



Addressing Table

Device	Interface	IP Address	IPv6 Address	IPv6 Link Local
R1	G0/0/1	192.168.1.1/24	2001:db8:acad:1000::1/64	fe80::1:1
D1	VLAN 1	192.168.1.2/24	2001:db8:acad:1000::2/64	fe80::d1:1
A1	VLAN 1	192.168.1.3/24	2001:db8:acad:1000::3/64	fe80::a1:1
PC1 (SNMP / Syslog service)	NIC	192.168.1.50/24	2001:db8:acad:1000::50/64	EUI-64

Objectives

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

Part 2: Configure and Verify SNMP

Part 3: Configure and Verify Syslog

Background / Scenario

Network Monitoring is critical to security and troubleshooting tasks. As your network grows and evolves, centralized monitoring becomes even more important. SNMP is a protocol that allows you to remotely monitor a wide range of settings and counters, be alerted when there are changes, and even remotely make configuration changes. Syslog is the log collector protocol. All of your devices should use Syslog to report device activity to a central location for correlation and records keeping. In this lab, you will configure both of these extremely important protocols.

Note: This lab is an exercise in configuring options available for SNMP and Syslog and does not necessarily reflect network troubleshooting best practices.

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Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.05c (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 (universalk9 image) and Cisco Catalyst 9200L with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Note: The default Switch Database Manager (SDM) template on a switch, such as Catalyst 2960, does not support IPv6. You must change the default SDM template to the dual-ipv4-and-ipv6 default template using the **sdm prefer dual-ipv4-and-ipv6 default** global configuration command. Changing the template will require a reboot.

Required Resources

- 1 Router (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 1 Switch (Cisco 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 1 Switch (Cisco 9200L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 1 PC (Choice of operating system with a terminal emulation program and packet capture utility installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Part 1: Build the Network and Configure Basic Device Settings and Interface Addressing

In Part 1, you will set up the network topology and configure basic settings and interface addressing on routers.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each device.

a. Console into each device, enter global configuration mode, and apply the basic settings. The startup configurations for each device are provided below.

Router R1

```
hostname R1
no ip domain lookup
ipv6 unicast-routing
banner motd # R1, Implement SNMP and Syslog #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
```

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```
 login
 exit
interface g0/0/1
 ip address 192.168.1.1 255.255.255.0
 ipv6 address fe80::1:1 link-local
 ipv6 address 2001:db8:acad:1000::1/64
 no shutdown
 exit
ntp master 3
end
```

Switch D1

```
hostname D1
no ip domain lookup
ipv6 unicast-routing
banner motd # D1, Implement SNMP and Syslog #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 exit
interface vlan 1
 ip address 192.168.1.2 255.255.255.0
 ipv6 address fe80::d1:1 link-local
 ipv6 address 2001:db8:acad:1000::2/64
 no shutdown
 exit
ip default-gateway 192.168.1.1
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
interface g1/0/23
 spanning-tree portfast
 switchport mode access
 no shutdown
 exit
interface g1/0/11
 spanning-tree portfast
 switchport mode access
 no shutdown
 exit
interface range g1/0/5-6
```

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```
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
 exit
ntp server 192.168.1.1
end
```

Switch A1

```
hostname A1
no ip domain lookup
ipv6 unicast-routing
banner motd # A1, Implement SNMP and Syslog #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 exec-timeout 0 0
 password cisco123
 login
 exit
interface vlan 1
 ip address 192.168.1.3 255.255.255.0
 ipv6 address fe80::a1:1 link-local
 ipv6 address 2001:db8:acad:1000::3/64
 no shutdown
 exit
ip default-gateway 192.168.1.1
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
interface range g1/0/1-2
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
 exit
ntp server 192.168.1.1
end
```

b. Set the clock on each device to UTC time.

c. Save the running configuration to startup-config.

d. Configure IPv4 and IPv6 addresses on host PC1 as shown in the addressing table.

e. Verify that R1, D1, and A1 can successfully ping 192.168.1.50.

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Part 2: Configure and Verify SNMP

The Simple Network Management Protocol (SNMP) is an application layer protocol that facilitates the exchange of management information between an agent and a management server. SNMP enables network administrators to monitor and manage network performance, find and solve network problems, and plan for network growth. SNMP management workstations can ask (get) for the value of a specific object identifier (OID) from the management information base (MIB) maintained by SNMP agents. The Manager can also configure (set) specific variable values in an OID. Additionally, the agent can send notifications (traps or informs) when an event occurs, or threshold is reached (an inform is a trap that must be acknowledged by the manager). Like any powerful tool, SNMP can be dangerous if not used properly, and securing the protocol and its uses are critical.

There are three SNMP versions. SNMPv3 is considered the most secure because it offers authentication and encryption, where SNMP versions 1 and 2 offer neither. SNMP access can also be limited using an access control list. SNMPv3 is rather complex to configure, and adoption is not universal. For this lab, we will configure SNMPv2c.

Step 1: Configure access-lists for SNMP.

Configure an access list on each device. This ACL will be used to specify exactly where SNMP get and set messages should be coming from. In this lab, the 192.168.1.0/24 network is the management network, and the SNMP manager is located at 192.168.1.50. **Configure this ACL on all three devices**:

```
R1(config)# ip access-list standard NMS-SERVER
R1(config-std-nacl)# permit host 192.168.1.50
R1(config-std-nacl)# exit
```

Step 2: Configure general SNMP information.

Configure general values to identify the device, its location, and a point of contact. **Configure this with appropriate values on all three devices**:

```
R1(config)# snmp-server location D1 Rack 1
R1(config)# snmp-server contact Student 555-1213
R1(config)# snmp-server chassis-id Cisco Device D1
```

Step 3: Configure SNMP community string.

SNMPv2c using a community string-based authentication. Access can be limited further by using an access list. Create a read-only community named CCNP that is limited by the NMS-SERVER ACL. **Configure this on all three devices**:

```
R1(config)# snmp-server community CCNP ro NMS-SERVER
R1(config)# snmp-server community CCNP rw NMS-SERVER
```

Step 4: Configure SNMP trap receiver.

Configure the NMS server that traps will be sent to. As a part of this command, specific traps or sets of traps to send can be specified. If no traps are specified, this receiver will be forwarded to all traps that are enabled. This particular configuration needs to be coordinated with the network management system and network monitoring requirements for the organization.

Configure 192.168.1.50 as a trap receiver using SNMPv2c and the community CCNP. **Configure this on all three devices**:

```
R1(config)# snmp-server host 192.168.1.50 version 2c CCNP
```

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Step 5: Configure interface index persistence.

Network monitoring systems record throughput and other interface statistics using SNMP polling. Each interface is referenced by its unique index number, which is dynamically assigned by the IOS during bootup. The index of each interface can be determined with the command **show snmp mib ifmib ifindex**. The dynamic assignment aspect of this can be problematic for documentation. Therefore, it is a good idea to instruct the system to keep a persistent list of interfaces, rather than a dynamic one. The use of this command creates a file stored in NVRAM. **Configure this on all three devices**:

```
R1(config)# snmp-server ifindex persist
```

Step 6: Run Wireshark on PC1.

Before enabling traps to be sent, run Wireshark or another packet capture utility on PC1 and filter the output to display only SNMP packets. This ensures that the packets are actually getting to the SNMP management server.

Step 7: Enable SNMP trap sending.

This final command actually enables the forwarding of traps to the configured trap receivers. As a part of this command, traps can be limited (as they can be in the **snmp-server host** command). Coordinate this with the network management system and network monitoring requirements for the organization. For this lab, you will simply enable all traps to be sent. **Configure this on all three devices**:

```
R1(config)# snmp-server enable traps
```

Step 8: Verify SNMP configuration.

a. To verify that traps are being sent, issue the command **debug snmp packets** and then enter and exit configuration mode on each device. You should see debug output indicating that a packet was sent each time. It might take a few minutes for each device to start sending traps.

```
R1# debug snmp packets
SNMP packet debugging is on
R1# config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
Oct 23 20:14:18.435: SNMP: Queuing packet to 192.168.1.50
Oct 23 20:14:18.499: SNMP: V2 Trap, reqid 1, errstat 0, erridx 0
 sysUpTime.0 = 704101
 snmpTrapOID.0 = ccmCLIRunningConfigChanged
 ccmHistoryRunningLastChanged.0 = 21365
 ccmHistoryEventTerminalType.1 = 3
 ccmHistoryEventTerminalUser.1 =
Oct 23 20:14:18.530: SNMP: Queuing packet to 192.168.1.50
Oct 23 20:14:18.530: SNMP: V2 Trap, reqid 2, errstat 0, erridx 0
 sysUpTime.0 = 704119
 snmpTrapOID.0 = ciscoConfigManEvent
 ccmHistoryEventCommandSource.2 = 1
 ccmHistoryEventConfigSource.2 = 2
 ccmHistoryEventConfigDestination.2 = 3
 ccmHistoryEventTerminalUser.2 =
R1(config)#
Oct 23 20:14:18.685: SNMP: Packet sent via UDP to 192.168.1.50
Oct 23 20:14:18.935: SNMP: Packet sent via UDP to 192.168.1.50
```

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```
R1(config)# exit
R1#
Oct 23 20:15:27.004: %SYS-5-CONFIG_I: Configured from console by console
R1#
Oct 23 20:15:27.004: SNMP: Queuing packet to 192.168.1.50
Oct 23 20:15:27.004: SNMP: V2 Trap, reqid 3, errstat 0, erridx 0
 sysUpTime.0 = 710966
 snmpTrapOID.0 = ccmCLIRunningConfigChanged
 ccmHistoryRunningLastChanged.0 = 702293
 ccmHistoryEventTerminalType.2 = 3
 ccmHistoryEventTerminalUser.2 =
Oct 23 20:15:27.254: SNMP: Packet sent via UDP to 192.168.1.50
R1#
```

b. Check the Wireshark output on PC1 and you should see the received SNMP messages in the packet capture.

c. Issue the **undebug all** command on each device to turn off the debugs.

Part 3: Configure and Verify Syslog

For a multitude of reasons, logging is a critical part of your network management plan. Cisco devices log to three general facilities: the console, the logging buffer, and a syslog server, if configured. All three of these can be controlled and configured so that the type of log message they record is specific. You have already experienced console logging simply by viewing the messages the device shows you when something happens. The logging buffer has also been collecting that same information. Both of these are local to the device. What you have not done yet is customize those facilities, nor have you configured and used a centralized syslog server, which would collect log messages from each of your devices and keep them so that you can examine and correlate events between different devices.

Before configuring logging, it is important that your devices have synchronized with an NTP server, so that they are all on the same time. This makes sorting and correlating events possible.

You also must have some kind of plan of how to separate and manage the log messages. Your plan must answer the questions “What do we do with all these logs?” and “What messages go where?”

Syslog messages are separated into eight different severity levels, numbered 0 through 7. The lower numbers indicate a more critical message. The severity levels also have keywords:

Severity Level	Keyword	Meaning
0	emergencies	System is unusable
1	alerts	Immediate action required
2	critical	Critical conditions
3	errors	Error conditions
4	warnings	Warning conditions
5	notifications	Normal but significant condition
6	informational	Informational messages
7	debugging	Debugging messages

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Note: When you designate a particular severity number as the specific message you want to log, you get that and anything with a smaller severity number. For example, if you set the level to 4, or use the keyword **warnings**, you capture messages with severity levels 4, 3, 2, 1, and 0.

Before we start configuring logging, let's look at how logging is configured by default.

```
R1# show run all | include logging
no service pt-vty-logging
no logging discriminator
logging exception 4096
no logging count
no logging message-counter log
no logging message-counter debug
logging message-counter syslog
no logging snmp-authfail
no logging userinfo
logging buginf
logging queue-limit 1024
logging queue-limit esm 0
logging queue-limit trap 1024
logging buffered 102400 debugging
logging reload message-limit 1000 notifications
no logging persistent
logging rate-limit console 40 except errors
no logging console guaranteed
logging console debugging
logging monitor debugging
logging cns-events informational
logging on
ethernet cfm logging alarm ieee
ethernet cfm logging alarm cisco
ethernet cfm logging ais
ethernet cfm logging lck
ip dhcp conflict logging
no device-tracking logging packet drop
no device-tracking logging theft
no device-tracking logging resolution-veto
no authentication logging verbose
no access-session wireless event-logging enable session-limit 0 event-limit 0
no access-session event-logging enable session-limit 0 event-limit 0
no mab logging verbose
no cts logging verbose
no dot1x logging verbose
 logging event link-status
 logging event link-status
 logging event link-status
 logging event link-status
 logging event link-status
 logging event link-status
ip ssh logging events
logging esm config
```

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```
logging dmvpn rate-limit 600
logging history size 1
logging history warnings
no logging alarm
logging trap informational
logging delimiter tcp
no logging origin-id
logging facility local7
no logging source-interface
logging server-arp
snmp-server enable logging setop
 logging synchronous
no device-tracking binding logging
netconf-yang cisco-ia logging ciaauthd-log-level error
netconf-yang cisco-ia logging confd-log-level error
netconf-yang cisco-ia logging nes-log-level error
netconf-yang cisco-ia logging onep-log-level error
netconf-yang cisco-ia logging sync-log-level error
```

Focusing on the lines in the output that are highlighted, we see that the router is configured to send debugging (or level 7) messages to the console, monitor, and buffer. We further see that logging is turned on, and that the trap logging level is informational (or level 6), and that the logging facility is number 7. The logging trap and logging facility commands have to deal with what messages are sent to an external server and how the server routes the log messages when they are received (Log facility 7 indicates one of several custom logging facilities, which are typically tied to a specific file).

Step 1: Modify buffered logging.

The logging buffer is set to hold 4096 bytes in a circular buffer and keep log messages at the debugging level and below. 4096 bytes is not quite enough space for a busy system, so you need to change the log buffer size to something larger. We will not be sending debugging messages to the syslog server, so the buffer is the only place those messages are stored. We will leave the logging buffer level at debugging for now and set the size of the buffer to 16384 bytes. **Configure this on all three devices**.

```
R1(config)# logging buffered 16384
```

Step 2: Modify the logging trap level.

The default logging trap level is 7 (keyword: debugging), and we do not really want debug messages in the logs that we will have to archive, so we need to change the logging trap level to 6 (keyword: informational). **Configure this on all three devices**.

```
R1(config)# logging trap informational
```

Step 3: Configure the Syslog server.

Next you need to configure the host address for the syslog server. In this lab, the Syslog server is 192.168.1.50. You should get a message saying logging to 192.168.1.50 has started. Configure this on all three devices:

```
R1(config)# logging host 192.168.1.50
R1(config)#
*Jan 30 19:35:58.039: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.50 port
514 started - CLI initiated
```

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Step 4: Verify the device configuration.

Check the device configuration by issuing the command **show logging** at the privileged exec prompt.

```
R1# show logging
Syslog logging: enabled (0 messages dropped, 2 messages rate-limited, 0 flushes, 0
overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

    Console logging: level debugging, 63 messages logged, xml disabled,
                     filtering disabled
    Monitor logging: level debugging, 0 messages logged, xml disabled,
                     filtering disabled
    Buffer logging:  level debugging, 5 messages logged, xml disabled,
                    filtering disabled
    Exception Logging: size (4096 bytes)
    Count and timestamp logging messages: disabled
    Persistent logging: disabled

No active filter modules.

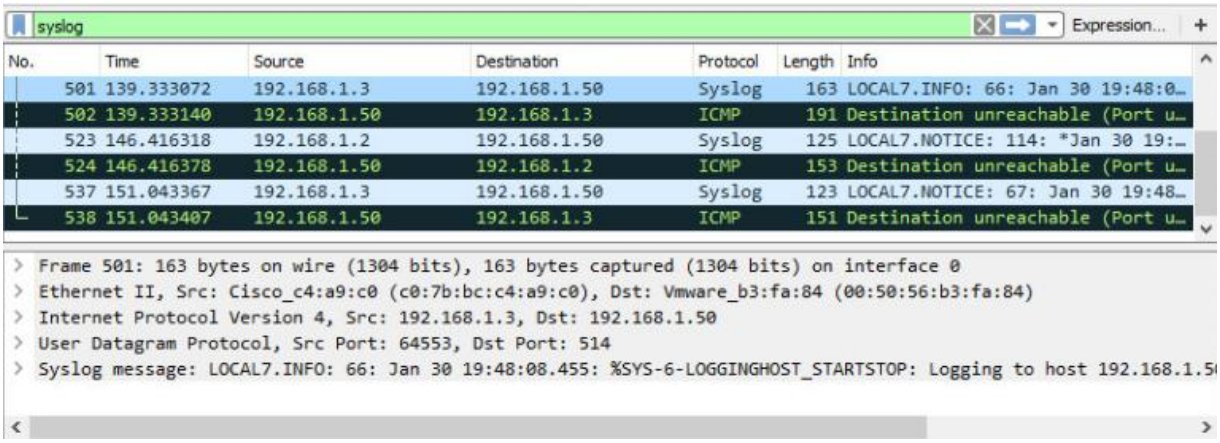
    Trap logging: level informational, 68 message lines logged
        Logging to 192.168.1.50  (udp port 514, audit disabled,
              link up),
              3 message lines logged,
              0 message lines rate-limited,
              0 message lines dropped-by-MD,
              xml disabled, sequence number disabled
              filtering disabled
        Logging Source-Interface:       VRF Name:

Log Buffer (16384 bytes):

*Jan 30 19:25:12.331: %SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml
disabled, filtering disabled, size (16384)
*Jan 30 19:35:57.038: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.50 port
0 CLI Request Triggered
*Jan 30 19:35:58.039: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.50 port
514 started - CLI initiated
Jan 30 19:36:49.443: %PKI-6-AUTHORITATIVE_CLOCK: The system clock has been set.
Jan 30 19:37:58.857: %SYS-5-CONFIG_I: Configured from console by console
<output omitted>
```

Now go to PC1 and reset Wireshark (if you have not already) and change the filter from snmp to syslog. Go into and out of configuration mode on each device, and you should see syslog messages in Wireshark:

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Note: The Destination Unreachable messages are host 192.168.1.50, telling the devices that there is not actually a Syslog server running at this IP address.

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